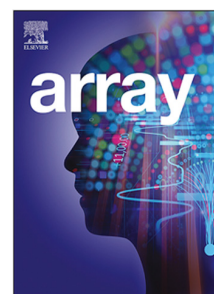


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ViMGS-SLAM: A real-time monocular 3DGS-based SLAM via multiscale vision transformers

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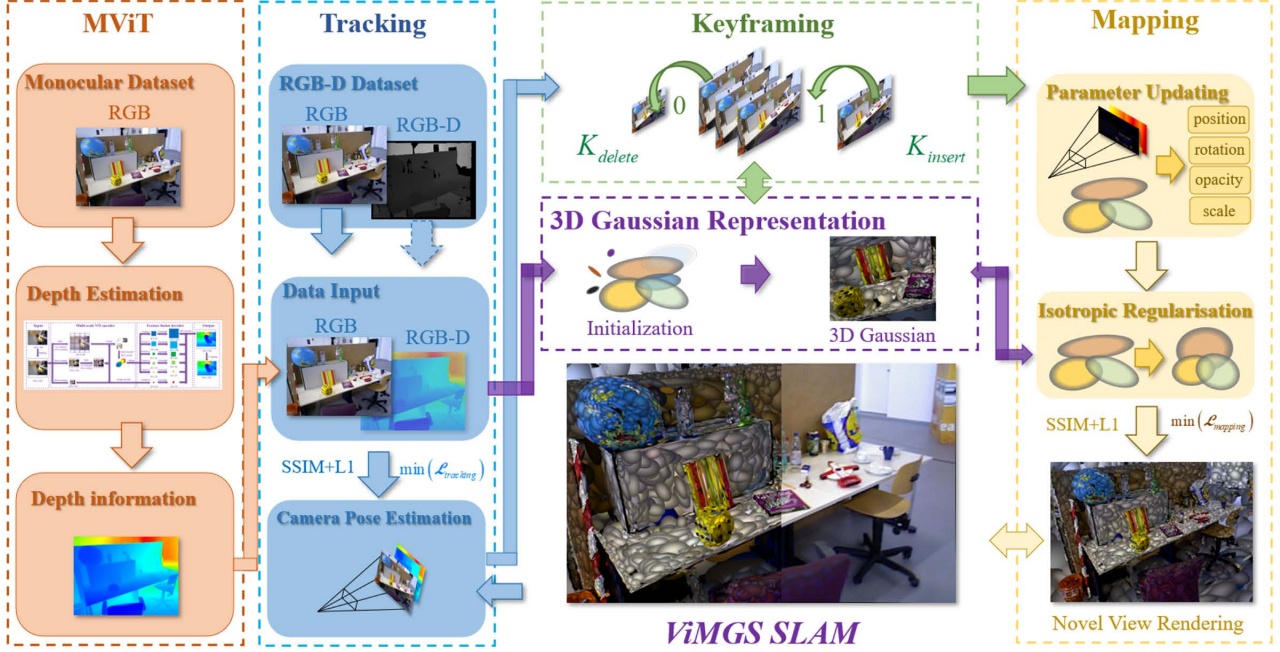
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Graphical Abstract

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Highlights

ViMGS-SLAM: A Real-Time Monocular 3DGS-based SLAM via Multiscale Vision Transformers

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- **A novel monocular 3DGS-based VSLAM fusion architecture with explicit metric scale recovery.** We present the first monocular SLAM framework that tightly integrates a Multi-scale Vision Transformer (MViT) with 3D Gaussian Splatting (3DGS). The MViT provides robust, metric-aware depth priors, while 3DGS enables high-fidelity real-time rendering and dense mapping, jointly resolving scale ambiguity and improving geometric consistency.
- **A multi-scale Vision Transformer for hierarchical geometric perception.** The proposed MViT extracts multiscale features via a three-level pyramid and constructs a five-level feature pyramid with shared-weight patch encoding and sliding-window merging. This design captures both fine local details and global context, significantly enhancing depth estimation accuracy in textureless and occluded regions.
- **Comprehensive experimental validation with state-of-the-art performance.** Extensive evaluations on TUM RGBD and Replica datasets demonstrate that ViMGSSLAM achieves an absolute trajectory error of 0.0236 m on TUM fr3 (46.0% improvement over MonoGS), rendering PSNR of 39.6 dB on Replica (+2.1 dB gain), and real-time operation (2.7 FPS monocular, 1130 FPS for rendering). The framework supports both monocular and RGBD inputs, and ablation studies confirm the contribution of each component.

ViMGS-SLAM: A Real-Time Monocular 3DGS-based SLAM via Multiscale Vision Transformers

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ABSTRACT

redVisual Simultaneous Localization and Mapping (VSLAM) is essential for autonomous systems, yet monocular implementations struggle with scale ambiguity and unreliable depth cues in textureless regions. We propose ViMGS-SLAM, a novel monocular framework that integrates a Multi-scale Vision Transformer (MViT) with 3D Gaussian Splatting (3DGS) to achieve real-time, metric-scale dense reconstruction. The MViT generates robust depth priors through a hierarchical pyramid architecture (three input scales, five feature levels), which are then used to initialize and constrain an explicit 3D Gaussian scene representation. A synchronous trackingmapping pipeline with adaptive keyframe selection and anisotropic regularization jointly optimizes camera poses and Gaussian parameters. On the TUM RGB-D fr3 dataset, ViMGS-SLAM reduces absolute trajectory error by 46.0% (from 0.0437 m to 0.0236 m) compared to MonoGS. On Replica, it achieves state-of-the-art novel-view synthesis with PSNR 39.6 dB, SSIM 0.976, and LPIPS 0.042, outperforming both NeRF-based and 3DGS-based methods. The system operates at 2.7 FPS end-to-end in monocular mode, while the differentiable renderer alone reaches 1130 FPS, confirming its efficiency. Ablation studies validate the contributions of the MViT depth prior, the adaptive keyframing, and the regularization terms.

1. Introduction

Visual Simultaneous Localization and Mapping (VSLAM) serves as a fundamental technology for constructing dense metric 3D maps of complex environments, thereby providing the critical geometric foundation for autonomous systems [1]. Among its various implementations, monocular VSLAM presents a compelling yet challenging paradigm due to its reliance on a single camera, which introduces significant difficulties in recovering 3D geometry from 2D images. Accurate depth estimation remains a core challenge[2], evolving from supervised methodsconstrained by expensive annotations to more scalable self-supervised paradigms.

Self-supervised monocular depth estimation, which bypasses the requirement for expensive ground-truth depth by framing depth prediction as a view synthesis problem, has garnered substantial research interest. This paradigm leverages the principle of photometric consistency, jointly optimizing depth and pose estimation networks to minimize reprojection error across multiple views of a scene [3]. Despite notable progress, self-supervised monocular depth estimation remains fundamentally constrained by several geometric challenges: scale ambiguity, unreliable depth cues in textureless regions, and blurred object boundaries resulting from insufficient geometric priors. To address these limitations, Vision Transformers (ViTs) [4] have emerged as a promising direction, offering global self-attention mechanisms that more effectively model long-range dependencies

and integrate contextual information compared to conventional convolutional networks. Variants tailored for dense prediction tasks, such as Multi-scale Vision Transformers (MViT) [5] and shifted-window architectures [6], aim to better capture hierarchical features and fine-grained local details. These architectural innovations are theoretically valuable for geometric reasoning, as they can yield more robust feature representations in textureless regions and improve handling of occlusions through global context. Nevertheless, within the self-supervised learning paradigm, these advanced architectures continue to grapple with fundamental theoretical limitations. Despite their enhanced representational capacity, they still struggle to resolve inherent scale ambiguity and remain limited in capturing geometric structures in textureless or occluded regions. Consequently, while ViTs enhance representational capacity, their potential for achieving robust, metric-aware geometric understanding remains constrained under the self-supervised paradigm [7].

Concurrently, 3D Gaussian Splatting (3DGS) [8] has emerged as a powerful explicit scene representation method, offering new possibilities for novel view synthesis and scene reconstruction. Its explicit and differentiable nature enables high-fidelity, real-time rendering, making it particularly attractive for VSLAM systems that require both mapping and visualization. As demonstrated in Table 1, 3DGS-based methods consistently outperform NeRF-based approaches in rendering quality, achieving higher PSNR and SSIM scores while maintaining lower LPIPS, which highlights the superiority of explicit scene representations for photorealistic reconstruction in SLAM systems. Integrating 3DGS into monocular VSLAM introduces new demands and opportunities for the front-end perception module [9]. Notably, among the 3DGS-based methods listed in Table ??, only MonoGS supports monocular input, yet its performance

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Table 1

Rendering Results of Different SLAM Methods [13] on Replica [14] Dataset.

Method	Algorithm	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
NeRF-based	NICE-SLAM [15]	24.42	0.81	0.23
	Vox-Fusion [16]	24.41	0.80	0.24
	Co-SLAM [17]	30.24	0.94	0.25
3DGS-based	GS-SLAM [18]	34.27	0.975	0.082
	SplaTAM [19]	34.11	0.970	0.100
	MonoGS SLAM [20]	37.50	0.960	0.070
	RGBD GS-ICP SLAM [21]	35.93	0.962	0.066

Best results are highlighted as first (**bold orange**), second (**yellow**), and third (**green**).

still leaves room for improvement, motivating our work to enhance monocular capability. The lack of reliable depth cues leads to significant scale ambiguity during Gaussian initialization, while sparse observational geometry often results in incomplete reconstruction and accumulated pose drift. To tackle these issues, several innovative strategies have emerged: hybrid architectures that combine traditional direct visual odometry with Gaussian mapping [10], photometric confidence-aware Gaussian management [11], and multi-scale deep regularization to mitigate scale drift [12]. Specifically, accurate initialization and continuous optimization of the 3D Gaussian map necessitate robust geometric priors, which is precisely the kind of global, context-aware understanding that advanced ViT architectures are designed to provide. The theoretical synergy is compelling, as ViTs can deliver strong, globally-informed features to estimate stable depth and infer scene structure from a single view, thereby guiding the 3DGS representation towards metric consistency and mitigating scale drift. Conversely, 3DGS provides a concrete, optimizable 3D structure where these learned features can be grounded and refined through rendering-based losses. However, effectively unifying these two paradigms within a monocular SLAM system presents significant challenges. These include designing an architecture that seamlessly bridges the 2D hierarchical feature space of a ViT with the 3D volumetric representation of Gaussian splats, managing the computational complexity of both components to maintain real-time performance, and devising a joint optimization strategy that leverages both the photometric fidelity of 3DGS rendering and the robustness of ViTs to visual ambiguities. Addressing these challenges is crucial for advancing monocular dense SLAM beyond its current limitations.

To address these challenges, we present ViMGSSLAM, a novel monocular VSLAM framework that tightly integrates a Multiscale Vision Transformer (MViT) with 3D Gaussian Splatting (3DGS). The MViT acts as a geometric perception backbone, producing metricscale depth priors through a hierarchical pyramid architecture that captures both fine local details and global context. These depth priors initialize and constrain an explicit 3D Gaussian scene representation, which is then continuously optimized via a

synchronous trackingmapping pipeline. The tracking module leverages photometric and depth residuals for accurate camera pose estimation, while the mapping module refines Gaussian parameters under anisotropic regularization to preserve sharp boundaries and suppress artifacts. An adaptive keyframe selection mechanism balances reconstruction quality and computational efficiency, and a newly introduced adaptive interval adjustment further enhances robustness to scene complexity. The remainder of this paper is organized as follows. Section 2 reviews related work on monocular depth estimation, selfsupervised depth learning, and 3DGS-based SLAM. Section 3 details the proposed framework, including the MViT architecture, the 3DGS representation, and the tracking, keyframing, and mapping modules. Section 4 presents extensive experimental results on depth estimation, novelview rendering, 3D Gaussian reconstruction, trajectory accuracy, ablation studies, and runtime analysis. Section 5 concludes the paper and discusses limitations and future directions.

The principal contributions of this work are threefold:

- **A monocular 3DGS-based SLAM architecture with explicit metric scale recovery.** We propose the first framework that tightly couples an MViT depth prior with a differentiable 3DGS representation. The MViT provides metricscale inverse depth maps, which anchor the scene geometry during tracking and mapping, effectively eliminating scale drift without requiring external sensors or camera intrinsics.
- **A hierarchical multi-scale Vision Transformer for robust geometric perception.** The MViT employs a threelevel image pyramid (640 $\times$ 640, 320 $\times$ 320, 160 $\times$ 160) and a weightshared patch encoder to produce a fivelevel feature pyramid (resolutions from 320 $\times$ 320 to 20 $\times$ 20). This design captures both fine details and global context, enabling accurate depth inference in textureless and occluded regions. All architectural detailsincluding patch size, stride, kernel sizes, and activation functionsare explicitly provided to ensure reproducibility.
- **Comprehensive experimental validation and ablation analysis.** Extensive experiments on TUM RGBD

and Replica datasets demonstrate state-of-the-art performance in trajectory estimation (ATE 0.0236 m, 46.0% improvement), novel view rendering (PSNR 39.6 dB), and runtime (2.7 FPS end-to-end, 1130 FPS for rendering). Ablation studies quantify the contribution of each loss component and the adaptive keyframing mechanism. The systems' limitations, such as the absence of loop closure, are discussed, and directions for future work are identified.

2. Related Work

2.1. Monocular Depth Estimation

Based on deep learning, monocular depth estimation methods can be principally categorized into supervised, semi-supervised, and self-supervised paradigms. 1) Supervised methods establish direct image-to-depth mappings using pixel-wise ground truth annotations. Representative works include DCNNs [22] and VGG-based frameworks [23], which achieve high accuracy but require expensive depth annotations, limiting their scalability in real-world applications [24]. 2) Semi-supervised methods mitigate data constraints through consistency regularization and pseudo-labeling techniques. While reducing annotation dependency, these approaches remain susceptible to error propagation and domain shift between labeled and unlabeled data distributions [25], [26]. 3) Self-supervised methods have gained prominence by reformulating depth estimation as a view synthesis problem. These approaches leverage geometric constraints from stereo or monocular sequences, minimizing photometric discrepancies through image warping operations [24]. Recent advances include Dyna-MSDepth [27], which employs multi-scale spatial-semantic interactions to generate stable depth estimates in dynamic environments. Despite progress, self-supervised monocular depth estimation faces persistent challenges: (1) dynamic object motion violating static scene assumptions, (2) inherent scale ambiguity in monocular setups, and (3) escalating training complexity with multi-modal data [28]. Future research should therefore prioritize the development of efficient multi-scale architectures that achieve robust scale consistency.

2.2. Self-supervised depth estimation

Self-supervised depth estimation has garnered substantial research attention due to its label-free training paradigm. Nevertheless, two major challenges still persist. First, self-supervised frameworks inherently lack metric scale recovery, which necessitates auxiliary sensor data (e.g., inertial measurements or camera intrinsics) to establish physical scale references [29]. However, inconsistencies in camera configurations across training datasets hinder the development of unified scale-aware models. While mainstream datasets emphasize structured scenes, they remain limited in representing dynamic objects and extreme illumination variations [30]. Addressing scale invariance and offset robustness across heterogeneous datasets thus constitutes a critical research direction for self-supervised paradigms. Second, domain gaps between datasets significantly degrade model

generalizability. Recent advances in zero-shot cross-dataset generalization aim to bridge this gap by enabling metric-compatible predictions without dataset-specific fine-tuning. Pioneering works such as MegaDepth [31] and MiDaS [32] have demonstrated the feasibility of large-scale multi-dataset training, while OmniData [33] introduced systematic strategies for cross-domain adaptation. Emerging architectures leveraging Transformers [34], ViT [35] and diffusion models [36] exhibit enhanced capacity for cross-dataset feature disentanglement, achieving notable progress in scale-consistent depth transfer. Despite these advancements, limitations persist in three aspects: 1) suboptimal accuracy under occlusions and textureless regions, 2) sensitivity to camera parameter variations, 3) compromised inference efficiency in multi-scale frameworks. To address these challenges, we propose a hierarchical multi-scale ViT architecture that synergistically integrates local texture details and global contextual cues through three distinct resolution pathways.

2.3. 3DGS-based SLAM

Within VSLAM research, reconstructing metrically consistent 3D environments from monocular image sequences remains a fundamental challenge. Extensive studies have demonstrated that elliptical primitives outperform conventional point or line features in geometric localization accuracy due to their enhanced spatial constraints [37], [38]. A seminal advancement was made by Zwicker et al. [39], who introduced Elliptical Weighted Average (EWA) Splatting - a sophisticated rendering framework employing anisotropic Gaussian kernels. Their work proved that EWA Splatting achieves alias-free reconstruction while preserving sharp discontinuities in volumetric data, effectively handling non-spherical scattering effects. This methodology demonstrates exceptional versatility across regular, curvilinear, and unstructured volumetric datasets, establishing a unified framework for multi-modal data processing. Subsequent ellipse-based methods advanced dynamic scene understanding [40], yet practical adoption in SLAM remains constrained by reliance on precise calibration, multi-view constraints, and sparse observation sensitivity. The advent of 3DGS has revolutionized this paradigm by enabling unified optimization of geometry and appearance through differentiable volumetric rendering, significantly advancing SLAM reconstruction efficiency and visual quality [41]. Despite these advancements, monocular 3DGS SLAM implementations remain under-explored, particularly regarding scale consistency in depth estimation and computational complexity management. Our work addresses these limitations through three key innovations: 1) A hybrid depth prediction network combining monocular geometric priors with multi-scale Gaussian uncertainty modeling, 2) A ViT-gaussian fusion framework for photorealistic novel view synthesis, and 3) Adaptive algorithms enabling real-time performance on both RGB and RGB-D streams.

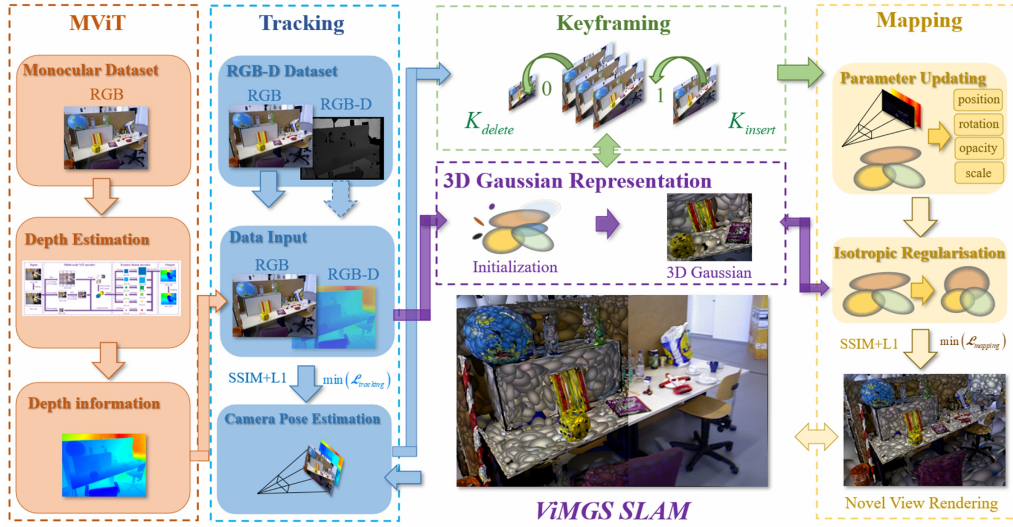


Figure 1: Overview of the ViMGS-SLAM system

3. Methods

3.1. System Overview

Fig. 1 illustrates the architectural framework of the ViMGS-SLAM system, which aims to achieve high-fidelity scene reconstruction and real-time rendering capabilities under complex illumination conditions. The system supports both RGB-D and RGB-only sensor configurations.

For monocular inputs, the MViT module (Section 3.2) generates dense depth priors through multi-scale attention mechanisms, ensuring geometrically consistent depth estimation without compromising temporal coherence. The system supports both RGBD and RGBonly sensor configurations. Following depth initialization or RGBD data acquisition, the 3DGS representation module (Section 3.3) transforms input data into anisotropic 3D Gaussian primitives via differentiable reparameterization. These primitives explicitly encode scene geometry (position, covariance) and photometric properties, forming a continuous volumetric representation suitable for differentiable rendering. Concurrently, the camera tracking module (Section 3.4.1) estimates camera poses in real time by iteratively minimizing a composite loss function that incorporates both L1 distance and structural similarity (SSIM). This formulation effectively balances subpixel intensity alignment with structural consistency, ensuring stable and accurate pose regression. The tracking output is then processed by the keyframing module (Section 3.4.2), which employs an adaptive selection strategy based on relative pose displacement and keyframe overlap density, complemented by an adaptive interval adjustment mechanism that dynamically modulates insertion frequency according to tracking loss. This strategy systematically prunes redundant keyframes to optimize computational efficiency while preserving the topological integrity of the constructed map. Within the mapping module (Section

3.4.3), ViMGSSLAM dynamically regulates the integration of 3D Gaussian primitives based on realtime pose estimations from the frontend tracking module, achieving iterative optimization of scene geometry parameters under isotropic regularization to preserve spatiotemporal consistency and suppress artifacts. Through this tightly coupled architecture, ViMGSSLAM maintains subcentimeter geometric accuracy and realtime performance, even in challenging scenarios involving textureless surfaces, occlusions, or repeated patterns.

3.2. Multi-scale Vision Transformers (MViT)

To enable geometrically accurate 3D reconstruction in visual SLAM and overcome the limitations of self-supervised monocular depth estimation, we propose a novel architecture based on MViT. This design integrates multi-scale visual analysis with geometric constraints in feature learning, addressing key challenges in self-supervised paradigms such as scale ambiguity, sensitivity to textureless regions, and blurred object boundaries.

As shown in Figure 2, the model comprises two tightly coupled components. First, the multi-scale input embedding stage employs linear projection and scale-adaptive positional encoding to map multi-resolution image patches into a unified feature space. All input images whether from TUM RGBD (originally 640×480) or other databases are padded to a square and resized to 640×640 . The MViT then constructs a three-level pyramid by further resizing to 640×640 , 320×320 , and 160×160 . At each scale, the image is divided into nonoverlapping patches using a convolutional layer with kernel size 16×16 and stride 16. Each patch is processed by a weight-shared Vision Transformer encoder, producing feature maps of size 24×24 per patch. These patch-level features are then merged into a full-resolution feature map at the corresponding pyramid level using a sliding-window

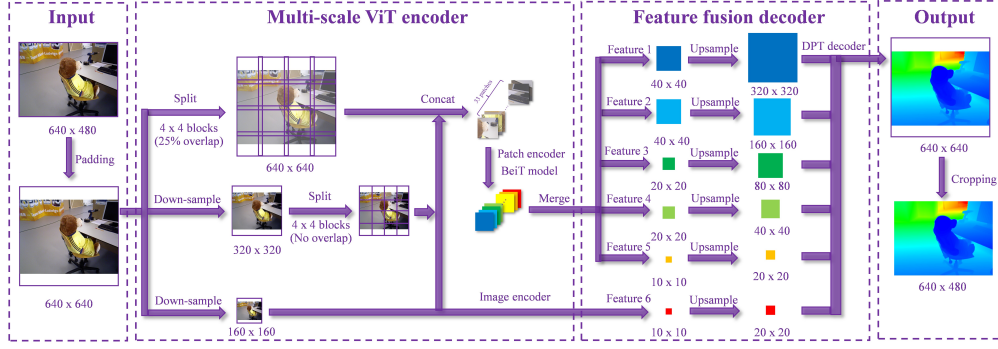


Figure 2: Schematic diagram of the MViT architecture integrated into the proposed framework. The experimental results (Section 4.2) illustrate its capability to effectively capture both global contextual information and localized spectral-spatial characteristics critical for robust monocular depth estimation, particularly in addressing the inherent limitations of self-supervised methods such as scale ambiguity, texture dependence, and boundary blurring.

merging strategy with 25% overlap, ensuring seamless reconstruction across patch boundaries. Subsequent feature projection employs 1×1 convolutions (stride 1) for channel alignment, spatial refinement uses 3×3 convolutions (stride 1), and upsampling stages rely on 2×2 transposed convolutions (stride 2) to progressively recover resolution. All convolutional layers in the feature fusion blocks and the output head use the ReLU activation function. This efficient multiscale design enables realtime operation under fast vehicle motion by keeping inference latency low. Second, the multi-scale feature fusion stage implements an attention-based feature pyramid network that ensures consistency between local geometry and global context through cross-scale interaction and spatialemantic alignment, enhancing depth discontinuity preservation and geometry inference in weakly-textured regions.

The multi-scale feature input and embedding module serves as the core encoding architecture of our MViT. It decomposes the input image into multi-scale patches $\mathbf{x}^{(s)}$ at scales $s \in \mathcal{S}$, applies linear projection via a learnable embedding matrix $\mathbf{E}$ together with scale-aware positional encodings $\mathbf{P}^{(s)}$, and extracts hierarchical representations using a pre-trained dual-branch Transformer encoder $\mathcal{H}^{(L)}$. Each patch is processed by the weight-shared encoder, producing a feature map of size 24×24 . These patch-level features are subsequently merged and upsampled to construct a multi-scale feature pyramid comprising five levels with spatial resolutions of 320×320 , 160×160 , 80×80 , 40×40 , and 20×20 , and channel dimensions of 256, 512, 1024, 1024, and 1024, respectively. Inspired by BeiT, the dual-branch encoder is designed to capture both local and global visual patterns: a local branch focuses on fine-grained details, while a global branch models long-range semantic dependencies. Through the fusion function Φ , which performs cross-scale concatenation and spatial-semantic alignment, these two streams collectively form a unified multi-scale feature pyramid. This design preserves both local geometric structures and global contextual understanding, establishing

an essential foundation for accurate geometric reconstruction in subsequent stages. The unified multi-scale feature representation $\mathcal{F}$ can be written as:

$$\mathcal{F} = \Phi \left(\left\{ \mathcal{H}^{(L)} \left(\mathbf{E} \cdot \mathbf{x}^{(s)} + \mathbf{P}^{(s)} \right) \right\}_{s \in \mathcal{S}} \right) \quad (1)$$

where the embedded representation for an input image at scale s is given by $\mathbf{E} \cdot \mathbf{x}^{(s)} + \mathbf{P}^{(s)}$. The token sequence $\mathbf{x}^{(s)}$ is linearly projected via $\mathbf{E}$ and augmented with scale-dependent trainable positional encodings $\mathbf{P}^{(s)}$. A Transformer encoder $\mathcal{H}^{(L)}(\cdot)$ with L layers then processes the embedded features, extracting rich contextual representations through multi-head self-attention and multilayer perceptron modules. The fusion function $\Phi(\cdot)_{s \in \mathcal{S}}$ integrates encoded features across all scales in $\mathcal{S}$ via cross-scale concatenation and spatialemantic alignment, yielding a unified multi-scale feature representation $\mathcal{F}$.

The dense feature upsampling and depth reconstruction module, built upon a DPT-based decoder, accurately maps multi-scale features to high-resolution depth estimates. After merging and upsampling, the obtained feature pyramid with five levels 320×320 , 160×160 , 80×80 , 40×40 , and 20×20 is first fused with global contextual features and then progressively upsampled via $2 \times$ transposed convolutions, ultimately restoring the original input resolution of 640×640 . Inter-scale misalignment is mitigated through explicit spatial alignment operations. The aligned features from all scales are then aggregated. The resulting unified multi-scale representation is condensed into a single-channel inverse depth map using a 1×1 convolutional layer. This design effectively integrates multi-scale cues in a coarse-to-fine manner, restoring high-resolution geometry while preserving pixel-level geometric continuity and sharp boundaries-forming the core of our sub-pixel geometric fidelity. The predicted inverse depth map $\hat{\mathbf{d}}^{-1}$ can be written as:

(4)

$$\hat{\mathbf{d}}^{-1} = \mathbf{W}_d \circ \text{Upsample}_{1:1} \left(\bigoplus_{s \in \mathcal{S}} \text{Align} \left(\text{Upsample}_{t \leftarrow s} (\mathbf{F}^{(s)} \oplus \mathbf{F}_g) \right) \right) \quad (2)$$

where $\hat{\mathbf{d}}^{-1}$ is the predicted inverse depth map, generated by applying a pointwise convolutional kernel $\mathbf{W}_d$ to the output of a final upsampling operation $\text{Upsample}_{1:1}$, which restores the aggregated multi-scale features to the original input resolution. This aggregation $\bigoplus_{s \in \mathcal{S}}$ combines features from all scales s in the set $\mathcal{S}$, where each scale's contribution is processed by a spatial alignment function Align that corrects feature misalignments. The input to the alignment function is the upsampled result of $\text{Upsample}_{t \leftarrow s}$, which rescales the fused features from scale s to a common target scale t . This fusion $\mathbf{F}^{(s)} \oplus \mathbf{F}_g$ combines scale-specific encoder features $\mathbf{F}^{(s)}$ with global contextual features $\mathbf{F}_g$ via a skip connection.

The MViT is trained as a zeroshot metric depth estimator on a diverse mixture of realworld and synthetic datasets using a combination of supervised losses, including mean absolute error (MAE) on metric ground truth and scale-and-shift-invariant losses on nonmetric data. This training enables the network to directly predict absolute scale inverse depth maps from a single RGB image without requiring camera intrinsics or metadata, providing a robust metric prior for subsequent SLAM modules.

3.3. 3DGS Representation

The 3DGS representation provides a unified parametric form for scene modeling that bridges continuous 3D geometry and discrete rendering through a collection of anisotropic Gaussians. The collection of anisotropic 3D Gaussians $\mathcal{G} = \{G_i\}_{i=1}^N$ is defined as:

$$\mathcal{G} = \{G_i(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i, O_i)\}_{i=1}^N, \quad \boldsymbol{\Sigma}_i = \mathbf{R}_i \text{diag}(\mathbf{S}_i) \mathbf{R}_i^\top \quad (3)$$

Where $\boldsymbol{\mu}_i \in \mathbb{R}^3$ is the Gaussian's center, $\boldsymbol{\Sigma}_i \in \mathbb{R}^{3 \times 3}$ is its covariance matrix, and $O_i \in \mathbb{R}$ is the opacity. The covariance is decomposed into a rotation matrix $\mathbf{R}_i \in \text{SO}(3)$ and scaling factors $\mathbf{S}_i \in \mathbb{R}^3$, ensuring positive semi-definiteness and enabling stable gradient-based optimization. This parametric form enables efficient differentiable optimization and direct splat-based rasterization.

During rendering, each 3D Gaussian is projected to 2D via a local affine approximation and composited using alpha blending. The projected 2D covariance $\tilde{\boldsymbol{\Sigma}}_i$ is derived from the Jacobian of the projective transform, while the final pixel color is obtained by depth-ordered blending of all contributing Gaussians based on their projected densities and opacities.

$$\mathbf{C}(p) = \sum_{i \in \mathcal{N}} \mathbf{c}_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j), \quad \alpha_i = O_i \exp\left(-\frac{1}{2} \boldsymbol{\Delta}_i^\top \tilde{\boldsymbol{\Sigma}}_i^{-1} \boldsymbol{\Delta}_i\right)$$

Where $\boldsymbol{\Delta}_i = p - \tilde{\boldsymbol{\mu}}_i$ is the pixel offset, $\tilde{\boldsymbol{\mu}}_i$ and $\tilde{\boldsymbol{\Sigma}}_i$ are the projected mean and covariance, α_i is the effective density, and $\mathbf{c}_i$ is the Gaussian's color. This formulation enables real-time, high-quality rendering through differentiable splatting and ordered blending.

3.4. Framework

Our framework comprises three tightly integrated components. Tracking estimates camera poses through photometric and depth residuals, keyframing maintains a representative subset of views for efficient mapping, and mapping optimizes the 3D Gaussian representation with geometric constraints.

These modules form a closed-loop system where tracking relies on keyframes for pose initialization, mapping utilizes keyframe observations for scene reconstruction, and keyframe selection is guided by both tracking stability and mapping quality requirements.

1) Tracking. Camera pose estimation is initialized via orthographic projection and optimized through a combined photometric and depth loss. The hybrid loss leverages both L1 and SSIM metrics for color alignment and incorporates depth residuals when available, with adaptive weighting between modalities. Gradient computation utilizes CUDA-based rasterization for efficiency, while Gaussian parameters remain fixed to ensure mapping stability. The tracking loss is formulated as:

$$\mathcal{L}_{\text{track}} = \lambda_c (\mathcal{L}_1(\hat{I}, I) + \mathcal{L}_{\text{SSIM}}(\hat{I}, I)) \odot M_{\text{rgb}} + \lambda_d \mathcal{L}_1(\hat{D}, D) \odot M_{\text{depth}} \quad (5)$$

Where $\hat{I}$ and $\hat{D}$ represent the rendered image and depth, I and D are ground-truth measurements, M_{rgb} and M_{depth} indicate valid pixels, and $\odot$ is element-wise multiplication. In monocular operation, where depth measurements are unavailable, we set $\lambda_d = 0$ and rely solely on photometric loss for pose estimation. However, to anchor scene geometry to a metric scale during mapping, the MViT predicted depth is treated as pseudo ground truth D , enforcing geometric consistency through the depth term $\mathcal{L}_{\text{depth}}$. Optimization runs for 50 iterations per frame using Adam to ensure convergence without map updates.

2) Keyframing. Keyframe management employs a sliding window strategy to balance reconstruction quality and computational load. It consists of two complementary components: a geometry-based selection criterion and an adaptive interval adjustment mechanism. New keyframes are inserted based on overlap and displacement thresholds relative to recent frames, while outdated ones are removed when overlap diminishes. This ensures sufficient viewpoint coverage while maintaining real-time operation. The keyframe selection criteria are defined as:

$$\text{Insert if: } \text{IoU}(\mathcal{G}_m, \mathcal{G}_n) < \theta \text{ or } \frac{L_{mn}}{\hat{D}_m} > \xi, \quad (6)$$

$$\text{Remove if: } \text{OC}(\mathcal{G}_m, \mathcal{G}_n) < \theta - \xi$$

where $\mathcal{G}_m, \mathcal{G}_n$ denote Gaussian representations, IoU and OC measure geometric overlap, L_{mn} is relative displacement, and $\hat{D}_m$ is median depth. Thresholds θ, ξ are dataset-specific (e.g., TUM: $\theta = 0.85, \xi = 0.05$; Replica: $\theta = 0.95, \xi = 0.04$). Monocular setups use static thresholds, which may limit adaptability.

To address this limitation, we introduce an adaptive keyframe interval adjustment mechanism that dynamically modulates the insertion frequency based on scene complexity. The keyframe interval Δ_{kf} is updated according to the tracking loss $\mathcal{L}_{\text{track}}$ as follows:

$$\Delta_{\text{kf}} = \begin{cases} \max(2, \Delta_{\text{default}} - 2), & \text{if } \mathcal{L}_{\text{track}} > \omega \\ \min(3, \Delta_{\text{default}} + 1), & \text{otherwise} \end{cases} \quad (7)$$

where Δ_{default} is the default interval and ω is a predefined loss threshold. The tracking loss $\mathcal{L}_{\text{track}}$ serves as a proxy for scene complexity and motion uncertainty. When the loss exceeds ω , indicating challenging conditions, the interval is reduced (subject to a minimum of 2) to insert keyframes more frequently, thereby strengthening geometric constraints and mitigating primitive distortion. Conversely, when the loss is low, the interval is increased (up to a maximum of 3) to reduce computational redundancy. This adaptive strategy enhances geometric consistency and textural authenticity while maintaining a favorable balance between resource usage and reconstruction quality.

3) Mapping. Scene reconstruction optimizes Gaussian parameters with explicit regularization to constrain anisotropic scaling in poorly observed regions. The mapping loss combines photometric, depth, and isotropic regularization terms, preventing artifact formation while preserving geometric fidelity under limited viewpoints. The mapping objective is defined as:

$$\mathcal{L}_{\text{map}} = \lambda_c \mathcal{L}_{\text{color}} + \lambda_d \mathcal{L}_{\text{depth}} + \lambda_{\text{reg}} \|\mathbf{S}_h - \bar{\mathbf{S}}_h \mathbf{1}\|_2 \quad (8)$$

Where $\mathcal{L}_{\text{color}}$ and $\mathcal{L}_{\text{depth}}$ mirror the tracking losses, $\mathbf{S}_h \in \mathbb{R}^3$ is the Gaussian scaling vector, $\bar{\mathbf{S}}_h$ is its mean, and $\mathbf{1}$ is a vector of ones. The regularization term penalizes deviation from isotropic scaling, suppressing excessive elongation. Weights satisfy $\lambda_c + \lambda_d = 1$ (with $\lambda_d = 0$ if depth is unavailable) and λ_{reg} controls regularization strength. Based on empirical validation, we set $\lambda_c = 0.8$, $\lambda_d = 0.2$ when depth is available, and $\lambda_{\text{reg}} = 0.05$.

Specifically, when operating in monocular mode, the depth prior D in $\mathcal{L}_{\text{depth}}$ is obtained from the MViT forward pass on the current keyframe. The L1 penalty between the rendered depth $\hat{D}$ and this prior anchors the optimization to a metric scale and effectively mitigates scale drift over time.

4. Experimental Results

4.1. Experiment setup

We evaluate our algorithms on two widely recognized benchmarks in the SLAM community: the Replica [14] and the TUM dataset [42], which comprises both monocular sequences and depthenhanced RGBD sequences. The TUM dataset exhibits inherent challenges in sensor acquisition, characterized by pronounced motion blur in color images and sparse depth measurements due to its Kinect v1 sensor limitations. These characteristics rigorously test an algorithm's capacity for detail preservation under realworld noise conditions. The Replica dataset provides synthetic environments with millimeteraccurate ground truth poses and dense depth maps, serving as an ideal benchmark for evaluating performance upper bounds in controlled scenarios.

All models are trained on an NVIDIA RTX 4090 GPU using PyTorch as the underlying deep learning framework. The training hyperparameters are as follows: the learning rates for the MViT encoder and decoder are set to 1.28×10^{-5} and 1.28×10^{-4} , respectively, with a cosine annealing schedule and a warmup phase of 1% of total epochs. The batch size is 128, and the Adam optimizer with a gradient clipping norm of 0.2 is employed. For RGB rendering quality assessment, we employ three widely adopted perceptual metrics: Peak SignaltoNoise Ratio (PSNR), Structural Similarity Index (SSIM), and Learned Perceptual Image Patch Similarity (LPIPS). In camera trajectory evaluation, the Root Mean Square Error (RMSE) of the Absolute Trajectory Error (ATE) is used to quantify positional accuracy.

To evaluate RGBonly performance, we benchmark our method against MonoGS [20], GOSLAM [43], and MGSO [44] on the TUM RGB dataset. For comprehensive analysis in RGBD scenarios, we compare with NeRFbased SLAM implementations (VoxFusion [16], NICESLAM [15], PointSLAM [45]) and 3D Gaussian Splattingbased systems (GSSLAM [18], SplatTAM [19], MonoGS [20]). All comparisons are conducted using both published benchmark results from the original papers and our reproduced experiments under identical hardware configurations with opensource implementations.

4.2. Depth Estimation Results

Fig. 3 compares the trajectory estimation accuracy of ViMGSSLAM and MonoGS [20] on the TUMRGBD fr3 dataset [42]. ViMGSSLAM achieves superior global alignment with an RMSE ATE of 0.0236 m, outperforming MonoGS (0.0437 m) by 46.0%. This improvement stems from the MViT architecture, whose hierarchical multiscale pyramid extracts both highfrequency local features and lowfrequency semantic information across varying resolutions, significantly reducing featurematching errors in occluded or texturepoor regions and suppressing cumulative drift. Transformerbased selfattention further establishes global contextual relationships, enabling spatialtemporal consistency where MonoGSs limited receptive fields lead to progressive drift. Keyframe analysis at frames 600, 1150, and 2400 further substantiates these advantages. At frame

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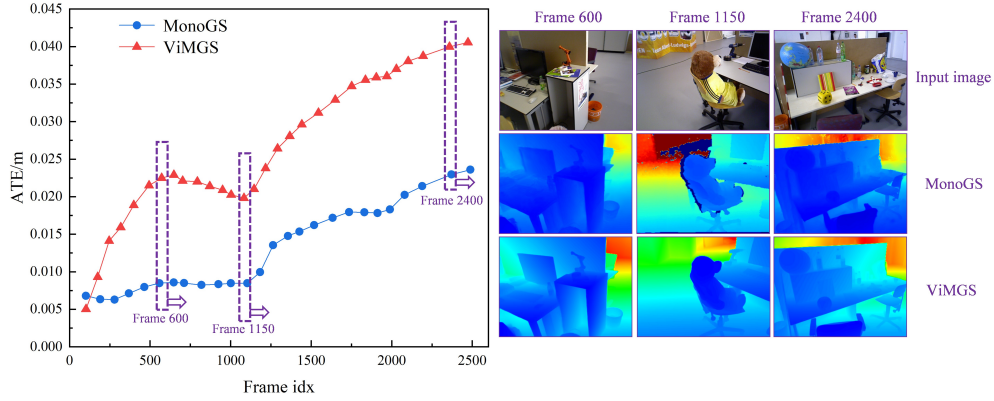


Figure 3: Visualization of ViMGS-SLAM for depth estimation on the TUM-RGBD fr3 dataset [42].

Table 2

Metrics Performance of ViMGS-SLAM on TUM-RGB and TUM-RGBD Datasets [42].

Method	Metric	RGB				RGB-D			
		fr1	fr2	fr3	Avg.	fr1	fr2	fr3	Avg.
Go-SLAM [17]	PSNR[dB]	11.71	14.81	13.57	13.36	15.79	16.12	16.50	16.14
	SSIM	0.41	0.44	0.48	0.44	0.53	0.53	0.57	0.54
	LPIPS	0.61	0.57	0.64	0.61	0.54	0.42	0.57	0.51
MGSO [44]	PSNR[dB]	18.07	24.10	21.61	21.26	—	—	—	—
	SSIM	0.66	0.80	0.75	0.74	—	—	—	—
	LPIPS	0.45	0.33	0.38	0.39	—	—	—	—
MonoGS [20]	PSNR[dB]	19.67	16.17	20.63	18.82	20.79	22.93	23.84	22.52
	SSIM	0.73	0.72	0.77	0.74	0.71	0.75	0.74	0.73
	LPIPS	0.33	0.31	0.34	0.33	0.31	0.22	0.23	0.25
ViMGS-SLAM	PSNR[dB]	21.28	25.14	23.38	23.27	23.56	25.06	25.30	24.60
	SSIM	0.74	0.80	0.80	0.78	0.79	0.81	0.84	0.81
	LPIPS	0.33	0.20	0.29	0.27	0.24	0.20	0.20	0.21

Best results are highlighted as first (bold orange), second (yellow), and third (green, not present in this table). Our method demonstrates superior performance over all baseline models in both monocular and RGB-D settings.

600, ViMGS-SLAM achieves an ATE of 0.68 cm versus 2.29 cm for MonoGS. At frame 1150, MonoGS exhibits structural discontinuities on planar surfaces and ambiguous object boundaries, while ViMGS-SLAM maintains accurate depth through multiscale feature fusion that combines edge details with semantic contours. By frame 2400, ViMGS-SLAM preserves temporal consistency via interframe feature propagation, dynamically aligning geometric and semantic information across frames. This mechanism enables robust tracking of fastmoving objects where MonoGS fails due to motionblurinduced depth drift.

Tab. 2 compares ViMGS-SLAM with several benchmark models on the TUM-RGB and TUM-RGBD datasets [42]. The TUM dataset, recorded by the Technical University of Munich's machine vision group, provides RGB-D sequences with ground truth data for evaluating VSLAM systems.

Under monocular input, ViMGS-SLAM outperforms GO-SLAM [17], MGSO [44], and MonoGS [20] across all validation metrics. ViMGS-SLAM achieves 23.27 dB (PSNR), 0.78 (SSIM), and 0.27 (LPIPS), surpassing the second-best method by +2.01 dB, +0.04, and -0.06, respectively. This highlights ViMGS-SLAM's superior depth estimation and feature extraction capabilities, particularly in dynamic scenes and weakly-textured regions. With depth-assisted input, ViMGS-SLAM maintains a leading edge with 24.60 dB (PSNR), 0.81 (SSIM), and 0.21 (LPIPS), exceeding the second-best method by +2.08 dB, +0.08, and -0.04. Notably, the TUM dataset contains challenging dynamic indoor scenes (e.g., moving objects, humans) and illumination variations, exacerbating depth estimation complexity. The performance gains stem from ViMGS-SLAM's MViT architecture, which captures global motion patterns via coarse-grained hierarchical feature fusion and enhances

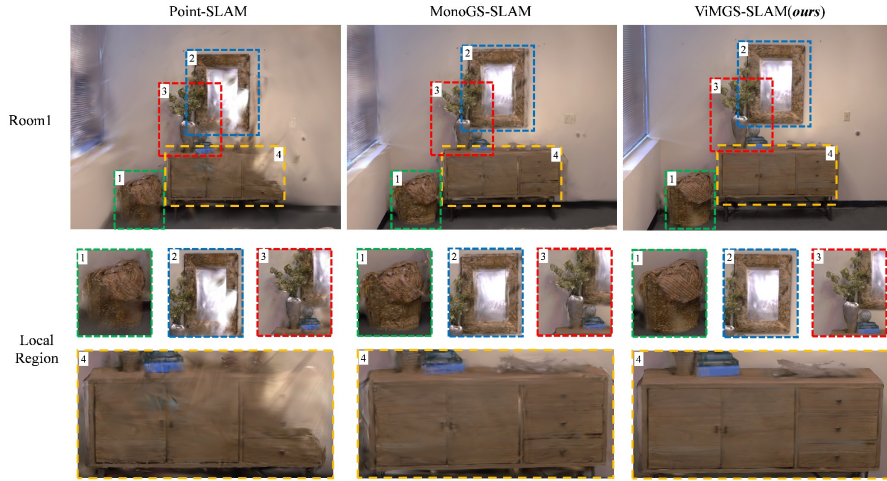


Figure 4: Rendering examples on Replica [14]. The proposed framework demonstrates enhanced photorealistic rendering fidelity through superior environmental context awareness and improved 3D reconstruction accuracy in complex geospatial scenarios.

local edge and texture perception through fine-grained attention mechanisms. Remarkably, ViMGS-SLAM’s monocular depth estimation accuracy on the TUM-RGB dataset closely approximates that of its depth-assisted counterpart on TUM-RGBD. This indicates that the MViT architecture achieves near-sensor-level precision in depth inference, significantly improving 3DGS-based scene reconstruction fidelity.

4.3. Novel View Rendering Results

Fig. 4 provides a comparative visualization of scene reconstruction quality among Point-SLAM [45], MonoGS [20], and the proposed ViMGS-SLAM on the Room1 sequence from the Replica dataset [14]. With identical RGB-D input, ViMGS-SLAM achieves superior geometric accuracy, particularly in preserving high-frequency details such as sharp edge contours and fine surface irregularities. This advantage stems from the MViT architecture, whose hierarchical multi-scale design integrates multi-resolution patch embedding, scale-aware positional encoding, and a dual-branch Transformer encoder to extract both local geometric cues and global context. As highlighted in Regions 1, 2, 3, and 4, ViMGS-SLAM maintains sub-centimeter structures (e.g., layered cloth folds in Region 1), produces semantically consistent completions in partially observed areas (Region 3), reduces artifacts at material boundaries (e.g., mirror surface in Region 2), and accurately recovers fine textures (e.g., wooden desktop in Region 4). In contrast, both Point-SLAM and MonoGS exhibit noticeable blurring, incomplete surfaces, or spurious textures in these challenging regions.

As quantitatively demonstrated in Table 3, Experimental results on the Replica benchmark [14] demonstrate that ViMGS-SLAM achieves favorable overall performance, validating its capability to deliver accurate reconstruction while

operating in real time. The results demonstrate ViMGS-SLAM’s comprehensive superiority across all metrics: 1) Radiometric accuracy. The proposed method achieves 39.60 dB (+1.69 dB vs. second-best). This result approaches the near-lossless reconstruction threshold (40 dB), indicating minimal deviation from ground truth measurements. 2) Structural integrity. With a structural similarity index of 0.976, our solution shows certain improvements (+0.001) over both Point-SLAM and GS-SLAM implementations. While not universally dominant in all scenarios, ViMGS-SLAM consistently ranks first in aggregate SSIM performance. 3) Perceptual fidelity. The learned perceptual image patch similarity metric reaches 0.042, representing a 66.7% reduction in perceptual discrepancies compared to the runner-up method.

These performance gains can be directly attributed to the novel architectural components introduced in our framework. First, the MViT contributes significantly to radiometric accuracy by capturing hierarchical features that preserve fine-grained intensity variations while suppressing noise through its global attention mechanisms. This enables the network to maintain geometric consistency and structural integrity across varying viewpoints and challenging geometric configurations. Second, the explicit 3D Gaussian Splatting representation ensures structural integrity by providing a physically-grounded geometric model that can accurately represent scene surfaces with anisotropic Gaussian kernels. The differentiable nature of this representation allows for precise optimization of scene geometry, leading to improved SSIM scores. Third, the synergistic integration of MViT features with the 3DGS pipeline enhances perceptual fidelity by enabling context-aware rendering that preserves high-frequency details and semantic coherence. The hierarchical feature aggregation in MViT provides rich contextual

Table 3

New View Rendering Results on Replica dataset [14]

Category	Method	Metric	Room0	Room1	Room2	Office0	Office1	Office2	Office3	Office4	Avg.
NeRF-based	Vox-Fusion [16]	PSNR[dB]	22.39	22.36	23.92	27.79	29.83	20.33	23.47	25.21	24.41
		SSIM	0.683	0.751	0.798	0.857	0.876	0.794	0.803	0.847	0.801
		LPIPS	0.303	0.269	0.234	0.241	0.184	0.243	0.213	0.199	0.236
	NICE-SLAM [15]	PSNR[dB]	22.12	22.47	24.52	29.07	30.34	19.66	22.23	24.92	24.42
		SSIM	0.689	0.757	0.814	0.874	0.886	0.797	0.801	0.856	0.809
		LPIPS	0.330	0.271	0.208	0.229	0.181	0.235	0.209	0.198	0.233
	Point-SLAM [45]	PSNR[dB]	32.4	34.08	35.5	38.26	39.16	33.99	33.48	33.49	35.17
		SSIM	0.974	0.977	0.982	0.983	0.986	0.96	0.96	0.979	0.975
		LPIPS	0.113	0.116	0.111	0.100	0.118	0.156	0.132	0.142	0.124
3DGS-based	SplaTAM [19]	PSNR[dB]	32.86	33.89	35.25	38.26	39.17	31.97	29.70	31.81	34.11
		SSIM	0.980	0.970	0.980	0.980	0.980	0.970	0.950	0.950	0.970
		LPIPS	0.070	0.100	0.080	0.090	0.090	0.100	0.120	0.150	0.100
	GS-SLAM [18]	PSNR[dB]	31.56	32.86	32.59	38.7	41.17	32.36	32.03	32.92	34.27
		SSIM	0.968	0.973	0.971	0.986	0.993	0.978	0.970	0.968	0.975
		LPIPS	0.094	0.075	0.093	0.050	0.033	0.094	0.110	0.112	0.082
	MonoGS [20]	PSNR[dB]	34.83	36.43	37.49	39.95	42.09	36.24	36.7	36.07	37.5
		SSIM	0.954	0.959	0.965	0.971	0.977	0.964	0.963	0.957	0.960
		LPIPS	0.068	0.076	0.075	0.072	0.055	0.078	0.065	0.099	0.070
	ViMGS-SLAM	PSNR[dB]	37.22	38.34	40.292	43.06	44.31	38.23	37.87	37.45	39.6
		SSIM	0.971	0.970	0.979	0.984	0.985	0.976	0.974	0.966	0.976
		LPIPS	0.041	0.047	0.0438	0.034	0.031	0.044	0.039	0.057	0.042

Note: Best results are highlighted as first (bold orange), second (yellow), and third (green). Our method demonstrates superior performance over all state-of-the-art dense prediction baseline models in terms of PSNR, SSIM, and LPIPS metrics.

information that guides the Gaussian optimization process, while the 3DGS representation ensures efficient and accurate rendering of these learned features.

Collectively, these innovations establish ViMGS-SLAM as an outstanding work in VSLAM systems, particularly excelling in environments requiring high-frequency detail preservation and real-time operational constraints. The framework's ability to simultaneously achieve superior radiometric accuracy, structural integrity, and perceptual fidelity demonstrates the effectiveness of combining transformer-based feature extraction with explicit differentiable scene representations for dense visual SLAM.

4.4. 3D Gaussian reconstruction

As shown in Fig. 5, we analyze the 3D Gaussian reconstruction processes between ViMGS and MonoGS [20] on the TUM RGB fr3 dataset [42]. The experimental results reveal that ViMGS-SLAM achieves significantly superior RMSE ATE accuracy compared to MonoGS. Notably, ViMGS-SLAM generates 95,156 Gaussian primitives in the proximity region, representing a 170.64% increase relative to MonoGS's 35,159 primitives. This substantial disparity in primitive density demonstrates ViMGS SLAMs capability to produce more geometrically complete point clouds while maintaining millimeter-level positioning accuracy.

Fig. 6 presents a qualitative comparison of Gaussian primitive generation between ViMGSSLAM and MonoGS [20] on the TUM RGBD fr2 dataset [42], with four local regions highlighted for detailed analysis. The overall distribution

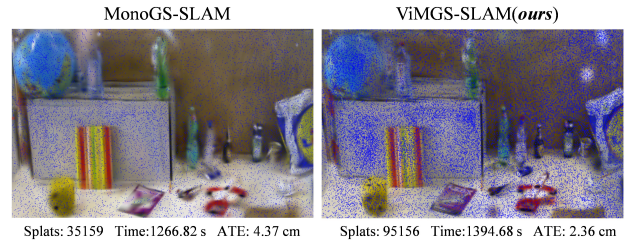


Figure 5: 3D Gaussian reconstruction process of ViMGS-SLAM and MonoGS on the TUM-RGB fr3 dataset [42].

of generated Gaussians indicates that ViMGSSLAM produces a significantly denser set of primitives compared to MonoGS, corroborating the quantitative findings in Fig. 5. Regions 1 and 4 demonstrate that ViMGSSLAM generates more refined and numerous Gaussians, leading to a denser and more detailed scene reconstruction. Moreover, the Gaussians produced by ViMGSSLAM exhibit a more isotropic (circular) shape, confirming the effectiveness of the anisotropic regularization term introduced in Equation 8. Regions 1 and 2 illustrate that ViMGSSLAM achieves more accurate color reconstruction, closely matching the original scene; this is attributed to the multiscale feature fusion and global context modeling of the MViT architecture, which enables precise texture and color perception. Region 3 reveals that ViMGSSLAM correctly reconstructs the display surface, whereas MonoGS erroneously generates light-colored Gaussians; this demonstrates that the depth prior and

geometric constraints in ViMGS-SLAM effectively prevent such artifacts by enforcing consistency with the observed geometry.

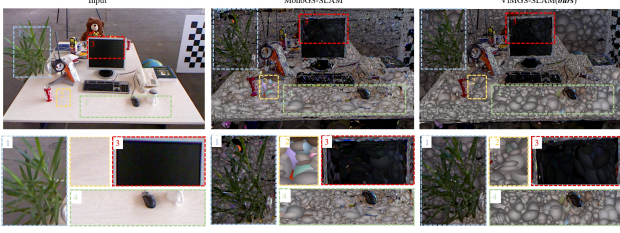


Figure 6: Qualitative comparison of Gaussian primitive generation between ViMGS-SLAM and MonoGS [20] on the TUM RGB-D fr2 dataset [42].

4.5. Trajectory estimation performances

Fig. 7 presents a comparative analysis of trajectory estimation performance between **ViMGS-SLAM** and MonoGS [20] on the TUM-RGB fr3 dataset [42]. The results demonstrate that **ViMGS-SLAM** maintains consistently lower RMSE ATE compared to MonoGS across the entire operational sequence. Both methods, however, exhibit transient error accumulation during the initialization and termination phases of the systems operation cycle, which may be attributed to sensor calibration convergence dynamics and boundary condition uncertainties. Meanwhile, both models exhibit significant error fluctuations during sharp turning maneuvers, a scenario inherently challenging for dynamic object tracking due to rapid viewpoint changes (particularly with moving occluders). However, **ViMGS-SLAM** demonstrates superior robustness, achieving a mean tracking error 50% lower than MonoGS. This performance gap underscores **ViMGS-SLAM**'s capability to mitigate error propagation through temporal feature aggregation and occlusion-aware attention.

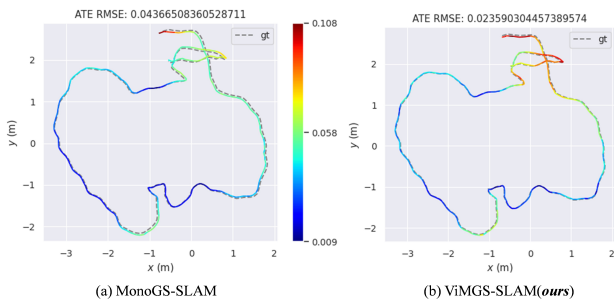


Figure 7: Comparative performance analysis of ViMGS and MonoGS [20] for trajectory estimation on TUM-RGB fr3 dataset [42].

Overall, the integration of the MViT depth prior with the explicit 3DGS representation enables ViMGS-SLAM to achieve robust metricscale reconstruction and accurate camera tracking. As demonstrated in Fig. 3, the MViT generates highfidelity depth maps with sharp boundaries, serving as

reliable geometric anchors throughout the pipeline. This leads to a substantial reduction in scale drift, as evidenced by the absolute trajectory error on the TUM fr3 sequence (Fig. 7), where ViMGS-SLAM outperforms MonoGS by 46.0%. Furthermore, Table 2 and Table 3 shows that ViMGS-SLAM consistently delivers superior rendering quality across all TUM and Replica sequences, achieving higher PSNR and SSIM scores and lower LPIPS values than the compared baselines. Collectively, these results validate the effectiveness of the proposed framework in enhancing both geometric consistency and photometric fidelity.

4.6. Ablation Experiments Results

Table 4 systematically evaluates the contributions of individual loss components through controlled ablation experiments on three TUM RGB sequences (fr1, fr2, fr3) [42], using RMSE ATE (in centimeters) as the primary metric. As described in Section 3.4, the ViMGS-SLAM framework employs photometric loss and depth loss for camera pose estimation during tracking, while incorporating photometric loss, depth loss, and isotropic regularization for scene reconstruction during mapping (see Equations 58). To investigate their respective roles, we conducted ablation studies on these three components, with results summarized below.

1) **Ablation of Photometric Loss.** When photometric loss is removed, the system relies solely on depth loss and isotropic regularization for both tracking and mapping. The RMSE ATE increases by 2.41 cm compared to the full model, underscoring the critical role of photometric constraints in maintaining localization accuracy. This degradation occurs because photometric loss provides essential visual feedback through pixelintensity alignment between current and reference frames, which is particularly crucial for handling illumination variations and textureless regions where depth cues are unreliable.

2) **Ablation of Depth Loss.** Removing the depth loss results in an RMSE ATE increase of 1.80 cm, highlighting its importance for geometric accuracy. The depth loss supplies metricscale constraints by leveraging scene geometry, which directly mitigates the scale ambiguity inherent in monocular SLAM. Nevertheless, the system retains functional tracking and mapping capabilities, indicating that photometric and regularization terms can partially compensate for the absence of depth supervision in static environments.

3) **Ablation of Isotropic Regularization.** The removal of isotropic regularization leads to a performance degradation comparable to that of depth loss ablation, with ATE increasing from 1.48 cm to 2.53 cm. This confirms its essential role in suppressing excessive Gaussian elongation and controlling error propagation. As discussed in Section 3.4, the regularization term $\|S_h - \bar{S}_h\|_2$ penalizes deviation from isotropic scaling, encouraging more compact and geometrically consistent primitives. The qualitative comparison in Fig. 6 further supports this finding, showing that ViMGS-SLAM produces more isotropic Gaussians than MonoGS, which directly contributes to improved reconstruction fidelity and trajectory accuracy.

Table 4

Ablation Experiments Results on TUM RGB dataset [42].

Method	TUM RGB			
	fr1	fr2	fr3	Avg.
w/o Color Loss	5.22	1.00	4.45	3.56
w/o Depth Loss	3.61	1.15	3.27	2.68
w/o Isotropic Loss	3.05	0.75	3.78	2.53
ViMGS-SLAM	1.73	0.36	2.36	1.48

Table 5

Runtime Results on TUM fr2 dataset [42]

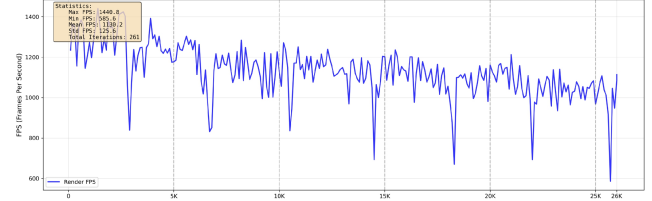
Input	Total Time [s]	FPS	Avg. Map. Iter.
Mono/tum/fr2	1325.9	2.7	19.6
RGBD/tum/fr2	1471.9	2.3	17.7

4.7. Runtime Results Analysis

Table 5 summarizes the endtoend runtime performance of ViMGSSLAM on the TUM fr2/desk sequence [42]. The system achieves an average frame rate of 2.7 FPS when processing monocular RGB streams and 2.3 FPS with RGBD input. These figures correspond to the complete synchronous execution of the pipeline per frame: tracking, keyframe management, and mapping are performed sequentially, guaranteeing that the map is consistently updated with the most recent observations. The mapping module averages 19.6 optimization iterations per keyframe in monocular mode and 17.7 iterations when depth data is available; this iteration count directly influences the overall frame rate, as each iteration involves gradient computation and parameter updates for the growing set of Gaussian primitives. The observed tradeoff between reconstruction fidelity and throughput reflects the inherent cost of dense geometry optimization in a synchronous design, while the rendering stage alone sustains over 1100 FPS, confirming the efficiency of the differentiable Gaussian splatting backend.

To better understand the computational characteristics, we examine the rendering subsystem separately. As shown in Fig. 8, the color refinement stage which involves only forward rendering of the 3D Gaussian representation achieves an average of 1130.2 FPS, with a peak of 1440.8 FPS. This high throughput demonstrates the exceptional efficiency of the differentiable Gaussian splatting pipeline. However, the overall frame rate remains lower because each tracking step requires multiple rendering passes to compute the photometric loss and gradients, and the mapping step iteratively updates Gaussian parameters. The computational cost of mapping scales with the number of Gaussians, which in our experiments grows to approximately 95,000 primitives on the TUM fr2 sequence. Consequently, the observed tradeoff between frame rate and reconstruction accuracy reflects the inherent cost of dense geometry optimization. While the current synchronous design prioritizes geometric consistency, future work could explore asynchronous execution

or adaptive iteration budgets to further increase throughput while maintaining quality.

**Figure 8:** Rendering frame rate during the color refinement stage on the TUM fr2 dataset [42].

While ViMGSSLAM achieves accurate dense reconstruction and realtime rendering, its current design does not incorporate explicit loop closure, relocalization after tracking failure, or longhorizon drift correction modules that are essential for a complete SLAM system. This is not a limitation specific to our approach; indeed, most existing 3DGSbased SLAM methods also focus on frontend tracking and dense mapping, with global optimization and loop detection remaining open research challenges in this emerging paradigm. Consequently, our system behaves as an odometryonly framework, and the absolute trajectory error gradually accumulates over extended sequences, as observed in Fig. 5 on the TUM fr2/desk dataset (2965 frames). In challenging scenarios such as rapid motion or severe occlusion, tracking loss may occur, and the system currently relies on reinitialization rather than active relocalization. Nevertheless, the adaptive keyframe selection and the interval adjustment mechanism provide some robustness to moderate motion changes and scene complexity. Integrating lightweight loop closure and relocalization modules into the 3DGS representation is a promising direction for future work, aiming to extend the framework to full SLAM capabilities while preserving its realtime performance.

5. Conclusion

In this paper, We presented ViMGS-SLAM, a realtime monocular VSLAM framework that synergistically integrates a Multiscale Vision Transformer (MViT) with 3D Gaussian Splatting (3DGS). The MViT generates metric-scale depth priors through a hierarchical pyramid architecture, which are used to initialize and constrain an explicit 3D Gaussian scene representation. A synchronous trackingmapping pipeline, equipped with adaptive keyframe selection and anisotropic regularization, jointly optimizes camera poses and Gaussian parameters. Extensive evaluations on the TUM RGBD and Replica benchmarks demonstrate that ViMGSSLAM significantly outperforms existing methods. On TUM fr3, it reduces the absolute trajectory error by 46.0% (from 0.0437m to 0.0236m) compared to MonoGS. On Replica, it achieves stateoftheart novelview synthesis with PSNR 39.6dB, SSIM 0.976, and LPIPS 0.042. The system operates at 2.7FPS endtoend in monocular mode, while

the differentiable renderer alone reaches 1130FPS, confirming its efficiency. Ablation studies validate the contribution of each component: the MViT depth prior, the photometric and depth losses, and the isotropic regularization term.

Despite these advances, ViMGSSLAM currently lacks explicit loop closure, relocalization, and longhorizon drift correction, which are common to most 3DGSbased SLAM systems. As shown in Fig. 5, trajectory error gradually accumulates over long sequences, highlighting the need for global optimization. Future work will focus on integrating lightweight loop closure modules and exploring asynchronous execution to further improve scalability and robustness. We believe this work provides a solid foundation for developing more sophisticated and efficient 3DGSbased SLAM systems in the future.

6. Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Huixin Zhu: Writing original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Junyang Zhao:** Writing review & editing, Resources, Supervision, Project administration, Funding acquisition. **Yutie Wang:** Writing original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Chen Xie:** Writing review & editing, Visualization, Validation, Software, Methodology, Formal analysis, Data curation. **Fan Zhang:** Conceptualization of this study, Methodology, Software. **Jiayi Wang:** Writing review & editing, Supervision, Project administration.

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Declaration of interests

☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: